

i-CAPE — Pattern of Questions, Syllabus and Marking Scheme

IMO · ISO · IEO · Grades 3 to 10

IMO — Innovative Maths Olympiad

Grade 3

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Number names (3-digit and 4-digit), comparing and forming numbers, and basic arithmetic operations (+, -, ×, ÷)

Measurement

Length, area, mass, volume, temperature, time, and money

Geometry

Shapes, patterns, and lines

Data Handling

Reading pictographs, tally marks, and bar graphs

Fractions

Understanding and identifying fractions

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen to mark their answers on the OMR sheet.
5. The use of electronic gadgets, including calculators, mobile phones, and smartwatches, is strictly prohibited.

Grade 4

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Number Sense

5-digit and 6-digit number names, comparing and forming numbers, basic arithmetic operations, factors and multiples

Arithmetic and Measurement

Fractions and decimals, time and money, and the measurement of length, area, mass, and volume

Geometry and Data

Plane geometry, symmetry, and data handling

Instructions

1. This question paper contains 35 compulsory questions.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks, totaling 70 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen on the OMR sheet.
5. Electronic devices such as calculators, mobile phones, smartwatches, and log tables are strictly prohibited.

Grade 5

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

7-digit and 8-digit numbers, patterns, and formation · Basic operations (+, -, ×, ÷), factors, and multiples

Fractions and Decimals

Fractions and decimals

Geometry

Geometry, shapes, and patterns

Measurement

Length, mass, volume, time, speed, and temperature · Money, perimeter, area, and volume

Data Handling

Data handling

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen for the OMR sheet.
5. Electronic devices such as calculators, mobile phones, smartwatches, and log tables are strictly prohibited.

Grade 6

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Knowing Our Numbers, Whole Numbers, and Playing with Numbers · Integers, Fractions and Decimals

Geometry

Basic Geometrical Ideas and Understanding Elementary Shapes · Symmetry

Mensuration and Data

Data Handling, Perimeter, and Area

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen on an OMR sheet.
5. Electronic devices such as calculators, mobile phones, or smartwatches are strictly prohibited.

Grade 7

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers and Algebra

Integers, Fractions and Decimals, and Rational Numbers · Exponents and Powers, and Algebraic Expressions · Simple Equations

Data Handling

Data Handling

Geometry and Mensuration

Lines and Angles, Triangles and Their Properties, Congruence of Triangles · Comparing Quantities, Perimeter and Area, and Visualising Solid Shapes

Instructions

1. This question paper contains 50 compulsory questions.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen for the OMR sheet.
5. Electronic devices, including calculators, mobile phones, smart watches, and log tables, are strictly prohibited.

Grade 8

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Rational Numbers, Playing with Numbers, and Exponents and Powers · Squares and Square Roots, Cubes and Cube Roots

Algebra

Algebraic Expressions, Identities, Factorisation, and Linear Equations in One Variable

Commercial Mathematics and Proportion

Direct and Inverse Proportion, Commercial Mathematics

Geometry and Mensuration

Visualising Solid Shapes and Mensuration

Instructions

1. This question paper contains 50 compulsory questions.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks, totaling 100 marks. There are no negative marks for incorrect answers.
4. Answers must be marked on an OMR sheet using a blue or black ball-point pen.
5. Electronic devices such as calculators, mobile phones, smartwatches, or log tables are strictly prohibited.

Grade 9

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Algebra and Number System

Number System and Polynomials · Coordinate Geometry and Linear Equations in Two Variables

Geometry

Euclid's Geometry, Lines and Angles, Triangles, Quadrilaterals, and Circles

Mensuration and Statistics

Heron's Formula, Surface Areas and Volumes, and Statistics

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.

Grade 10

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Algebra

Real Numbers, Polynomials, Pair of Linear Equations, and Quadratic Equations · Progressions

Geometry

Triangles and Coordinate Geometry · Circles

Trigonometry

Introduction to Trigonometry and Applications of Trigonometry

Mensuration and Data

Surface Areas and Volumes, Statistics, and Probability

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen to mark their answers on an OMR sheet.
5. Electronic devices, including calculators, mobile phones, and smartwatches, are strictly prohibited.

ISO — Innovative Science Olympiad

Grade 3

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Living World

Plants, Animals, and Birds · Food and Human Needs · Human Body

Physical Science

Matter and Materials · Light, Sound, and Force

Earth and Environment

Transport, Communication, and Safety Rules · Earth and Universe · Our Environment

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks, totaling 70 marks. There is no negative marking.
4. Students must use blue or black ball-point pens on OMR sheets.
5. Electronic gadgets, such as calculators or mobile phones, are strictly prohibited.

Grade 4

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Living World and Daily Life

Plants and animals, food, and clothes · Safety and first-aid, human body, and matter

Physical Science

Force, work, energy, air, and water

Earth and Environment

Earth, the universe, and the environment

Instructions

1. This question paper contains 35 compulsory questions.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks, totaling 70 marks. There are no negative marks for incorrect answers.
4. Candidates must use a blue or black ball-point pen to mark their answers on the OMR sheet.
5. Prohibited items include calculators, mobile phones, smartwatches, log tables, and other electronic gadgets.

Grade 5

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Matter and Earth Materials

Matter Around Us, Rocks and Minerals, and Air and Water

Biology and Health

Food and Health, Diseases, and the Human Body · Plants Around Us and Animals Around Us

Physics

Work and Energy, Force and Simple Machines

Earth and Environment

Sun, Moon and Earth, Pollution and Natural Disasters

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks, totaling 70 marks. There is no negative marking.
4. Students must use a blue or black ball-point pen for the OMR sheet.

5. Electronic devices such as calculators, mobile phones, or smartwatches are strictly prohibited.

Grade 6

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Biology

Components of Food · Diversity in the Living World and Living Creatures Characteristics

Chemistry

Materials Around Us and Methods of Separation · Air and Water

Physics and Earth

Motion, Measurement, and Temperature · Exploring Magnets, Light, and Beyond Earth

Instructions

1. This question paper contains 50 compulsory questions, totaling 100 marks.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marking.
4. Only blue or black ball-point pens are permitted for marking the OMR sheet.
5. The use of electronic gadgets such as calculators, mobile phones, or smartwatches is strictly prohibited.

Grade 7

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Biology

Nutrition in Plants and Animals, Respiration in Organisms, Transportation in Plants and Animals, and Reproduction in Plants

Chemistry

Acids, Bases, and Salts; Physical and Chemical Changes

Physics / General Science

Water as a Precious Resource, Heat, Motion and Time, Electric Current and Its Effects, and Light

Instructions

1. This question paper contains 50 questions, totaling 100 marks.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen to fill the OMR Answer Sheet.
5. Calculators, mobile phones, smartwatches, log tables, and other electronic gadgets are strictly forbidden.

Grade 8

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Biology

Cell structure and functions, microorganisms, and reproduction in animals · Crop production and management, and conservation of plants and animals

Chemistry

Coal, petroleum, combustion, and flame · Metals and non-metals

Physics

Force and pressure, friction, sound, light, and the chemical effects of electric current

Instructions

1. This question paper contains 50 questions, totaling 100 marks.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen for the OMR sheet.
5. Electronic devices such as calculators, mobile phones, and smartwatches are strictly prohibited.

Grade 9

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Chemistry

Matter in Our Surroundings and Is Matter Around Us Pure? · Atoms and Molecules and Structure of the Atom

Biology

The Fundamental Unit of Life, Tissues, and Improvement in Food Resources

Physics

Motion, Force and Laws of Motion, Gravitation, and Work and Energy

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Answers must be marked on an OMR sheet using a blue or black ball-point pen.
5. The use of calculators and electronic gadgets is strictly prohibited.

Grade 10

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Chemistry

Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, and Carbon and Its Compounds

Biology

Life Processes, Control and Coordination, Organism Reproduction, Heredity, and Our Environment

Physics

Light (Reflection and Refraction), The Human Eye and the Colourful World, Electricity, and Magnetic Effects of Electric Current

Instructions

1. This question paper contains 50 questions. All questions are mandatory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Candidates must use a blue or black ball-point pen to mark their OMR sheets.
5. Electronic gadgets such as calculators, mobile phones, smart watches, and log tables are strictly prohibited.

IEO — Innovative English Olympiad

Grade 3

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Word Meanings, Homonyms, Synonyms, Antonyms, Singulars, Plurals, Spellings, Homophones, Genders, Number, Proverbs, Idioms, Analogy · Parts of Speech – Nouns, Pronouns, Verbs, Adverbs, Adjectives, Prepositions, Articles, Tenses, Punctuations, Jumbled Words

Reading Comprehension

Read and interpret stories, anecdotes, and other text types · Understand information given in pictorial and time-table formats · Comprehend short texts and identify key details in messages, invitations, and similar formats

Interactive English

Ability to interpret and respond appropriately to functions such as apologies, greetings, introductions, conversations, and requests based on the situation

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.

Grade 4

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Word Meanings, Homonyms, Synonyms, Antonyms, Singulars, Plurals, Spellings, Homophones, Genders, Number, Proverbs, Idioms, Analogy · Parts of Speech – Nouns, Pronouns, Verbs, Adverbs, Adjectives, Prepositions, Conjunctions, Articles, Tenses, Punctuations, Jumbled Words · Basic Questions and Question Tags

Reading Comprehension

Read and interpret stories, anecdotes, and other text types · Understand information given in pictorial and time-table formats · Comprehend short texts and identify key details in messages, invitations, and similar formats

Interactive English

Ability to interpret and respond appropriately to functions such as apologies, greetings, introductions, conversations, and requests based on the situation, passing information, etc.

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.

Grade 5

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Word meanings, homonyms, synonyms, antonyms, singulars, plurals, spellings, homophones, genders, number, proverbs, idioms, analogy · Parts of speech – nouns, pronouns, verbs, adverbs, adjectives, prepositions, conjunctions, articles, punctuations, tenses, active voice and passive voice, subject – verb agreement, direct and indirect speech, etc.

Reading Comprehension

Read and interpret stories, anecdotes, and other text types · Understand information given in pictorial and time-table formats · Comprehend short texts and identify key details in messages, invitations, and similar formats

Interactive English

Ability to interpret and respond appropriately to functions such as apologies, greetings, introductions, conversations, and requests based on the situation, passing information, etc.

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.

Grades 6 & 7

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Parts of speech and their correct usage in the sentences, articles, tenses, punctuations, jumbled words, syllables, and word formation; vocabulary development; analogies; synonyms and antonyms; determiners, quantifiers, collocations; phrasal verbs; clauses and phrases; sentence structure; subject–verb agreement; identifying the odd-one-out; one-word substitutes; idiomatic expressions; direct and indirect speech; active and passive voice; etc.

Reading Comprehension

Understanding the story holistically and interpreting information in detail; comprehending phrases, phrasal verbs, and idioms both contextually and independently; analysing specific details and their implications; extracting information about events, characters, and references within the text; identifying underlying messages and moral perspectives; and related comprehension skills.

Interactive English

Using language for exchanging pleasantries; greeting; thanking; apologising; offering consolation; giving warnings; expressing opinions and advice; showing polite disagreement; introducing oneself; making requests; seeking permission; conveying information clearly and completely; and demonstrating proper etiquette in personal as well as telephonic conversations.

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.

Grades 8 & 9

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Parts of speech and their correct usage in the sentences, articles, tenses, punctuations, jumbled words, syllables, and word formation; vocabulary development; analogies; synonyms and antonyms; determiners, quantifiers, collocations; phrasal verbs; clauses and phrases; sentence structure; subject–verb agreement; identifying the odd-one-out; one-word substitutes; idiomatic expressions; direct and indirect speech; active and passive voice; etc.

Reading Comprehension

Understanding the story holistically and interpreting information in detail; comprehending phrases, phrasal verbs, and idioms both contextually and independently; analysing specific details and their implications; extracting information about events, characters, and references within the text; identifying underlying messages and moral perspectives; and related comprehension skills.

Interactive English

Formal written communication; expressing surprises, etiquette in personal and telephonic conversations; appropriate expressions; seeking permission; introducing oneself; ability to understand situation-based variations, conveying information clearly; giving advice; consoling; warning; engaging in arguments; and expressing polite disagreement.

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.

Grade 10

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Vocabulary and Functional Grammar

Parts of speech and their correct usage in the sentences, articles, tenses, punctuations, jumbled words, syllables, and word formation; vocabulary development; analogies; synonyms and antonyms; determiners, quantifiers, collocations; phrasal verbs; clauses and phrases; sentence structure; subject–verb agreement; identifying the odd-one-out; one-word substitutes; idiomatic expressions; direct and indirect speech; active and passive voice; etc.

Reading Comprehension

Understanding the story holistically and interpreting information in detail; comprehending phrases, phrasal verbs, and idioms both contextually and independently; analysing specific details and their implications; extracting information about events, characters, and references within the text; identifying underlying messages and moral perspectives; and related comprehension skills.

Interactive English

Using language for exchanging pleasantries; greeting; thanking; apologising; offering consolation; giving warnings; expressing opinions and advice; showing polite disagreement; introducing oneself; making requests; seeking permission; conveying information clearly and completely; and demonstrating proper etiquette in personal as well as telephonic conversations.

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.